

Description of Attributes

Mountain Lion Detection System

By: Fernando Gonzalez

Alerts

Stores every alert the controlling computer has received, split by age.

Attribute	Datatype	Description
recent	Vector<FiledReport>	Alerts received within the last 30 days, kept in full detail.
archive	Vector<FiledReport>	Alerts older than 30 days but received within the last year.

FiledReport

Pairs a received report with the ranger's classification of it.

Attribute	Datatype	Description
report	Report	The alert message received from the sensor.
status	String	The ranger's classification of the alert: "definite", "suspected", or "false".
ranger	*Ranger	Pointer to the ranger who classified the alert. Null until the alert is classified.

Report

The alert message sent by a sensor to the controlling computer.

Attribute	Datatype	Description
time	Int	Time the noise was detected or the alert was received (e.g., Unix timestamp in seconds).
sensorID	Int	The id of the Sensor that sent the alert.
noise	Noi	The detected noise: its type, strength, and location.

Noise

Describes a single detected animal noise.

Attribute	Datatype	Description
strength	Float	Strength of the detected noise.
type	Animal	The kind of animal the noise matches.
xPos	Float	East-west position of the noise, accurate to within 3 meters.
zPos	Float	North-south position of the noise, accurate to within 3 meters.

Sensor

A noise detection sensor placed in the park (approximates the Animals-R-Here device).

Attribute	Datatype	Description
outgoing	Report	The most recent report the sensor is sending to the controlling computer.
id	Int	Unique identifier of the sensor. Matches Report.sensorID.
targets	Vector<Animal>	Animal noise types this sensor is programmed to detect.

Animal

Represents a type of animal that a noise can match and that a sensor can be programmed to detect. This class has no attributes in the current design.

Alarm

Tracks the state of the alarm on the controlling computer.

Attribute	Datatype	Description
active	Bool	True while the alarm is sounding. Set to false when the ranger turns it off.
lastReport	Report	The report that most recently triggered the alarm. Used to decide whether a new noise is at a different location.

Ranger

A park ranger who uses the control program.

Attribute	Datatype	Description
name	String	Ranger's full name.
id	Int	Unique identifier of the ranger.